

1 Solution — GIAM 6.3

Problem 2

1.1 Problem

Define a relation $\square$ on the integers by

$$x \square y \iff x^2 = y^2.$$

Show that $\square$ is an equivalence relation. List the equivalence classes $x/\square$ for $0 \leq x \leq 5$.

1.2 Setup

Let $f : \mathbb{Z} \rightarrow \mathbb{Z}$ be the squaring map $f(n) = n^2$. Then

$$x \square y \iff f(x) = f(y),$$

so $\square$ is exactly equality on $\mathbb{Z}$ pulled back through f — the same recipe as the astrological-sign relation in 6.2 Problem 2 — and inherits all three equivalence properties from equality.

1.3 $\square$ Is an Equivalence Relation

- **Reflexive.** $x^2 = x^2$, so $x \square x$ for every $x \in \mathbb{Z}$.
- **Symmetric.** If $x \square y$, then $x^2 = y^2$, hence $y^2 = x^2$, so $y \square x$.
- **Transitive.** If $x \square y$ and $y \square z$, then $x^2 = y^2$ and $y^2 = z^2$, so $x^2 = z^2$, hence $x \square z$.

All three hold, so $\square$ is an equivalence relation. $\blacksquare$

1.4 Equivalence Classes — General Form

For each $x \in \mathbb{Z}$,

$$[x]_{\square} = \{y \in \mathbb{Z} : y^2 = x^2\} = \{y \in \mathbb{Z} : y = \pm x\} = \{x, -x\}.$$

Equivalently, $x \square y \iff |x| = |y|$, so a class is determined by the *absolute value* of any of its members. Each class has exactly two elements except $[0]_{\square} = \{0\}$, which has one (since $-0 = 0$).

1.5 The Classes for $0 \leq x \leq 5$

Applying $[x]_{\square} = \{x, -x\}$:

- $0/\square = [0]_{\square} = \{0\}$
- $1/\square = [1]_{\square} = \{1, -1\}$
- $2/\square = [2]_{\square} = \{2, -2\}$
- $3/\square = [3]_{\square} = \{3, -3\}$
- $4/\square = [4]_{\square} = \{4, -4\}$
- $5/\square = [5]_{\square} = \{5, -5\}$

These six classes are pairwise distinct because the squares $0, 1, 4, 9, 16, 25$ are six distinct values, so the fibers $f^{-1}(0), f^{-1}(1), \dots, f^{-1}(25)$ are pairwise disjoint.

1.6 Remark — The Full Quotient Set

Extending the listing beyond $x = 5$ produces no new classes once the absolute value has been seen — for instance $[-3]_{\square} = [3]_{\square}$. So the full quotient is

$$\mathbb{Z}/\square = \{[n]_{\square} : n \in \mathbb{Z}_{\geq 0}\} = \{\{0\}, \{\pm 1\}, \{\pm 2\}, \{\pm 3\}, \dots\},$$

and the map

$$[x]_{\square} \longmapsto |x|$$

is a bijection $\mathbb{Z}/\square \rightarrow \mathbb{Z}_{\geq 0}$. So $\square$ is, up to bijection, just the relation “have the same absolute value” — and the six classes listed above are simply the first six entries of this

enumeration.